MOTIONDRAW

A camera instrument. A guide line sits over the live camera image inside a white passepartout frame with a square grid. Whatever crosses the line plays a note. Position along the line picks the pitch, quantised to a scale, and the grid cell where it happened flashes black.

Live web build: https://berlogabob.github.io/MotionDraw/ (tap once to start audio, allow the camera). Runs natively on macOS, iOS and Android.

PLAYING

1. Point the camera at something that moves (people, traffic, rain on a window) or at a still scene with edges and spots (sunlight through blinds, a bookshelf).
2. Drag inside the frame to move the line. Double-tap flips it between vertical and horizontal. The line always snaps to a grid line.
3. Pick a mode:
   - DYNAMIC — the line stays put; anything that moves across it plays. Low notes are at the bottom (vertical line) or left (horizontal line); LOW flips that.
   - STATIC — press PLAY and the line sweeps across the frame as a playhead; every edge or bright spot it crosses plays. SWEEP sets the time for one pass. The sweep direction follows LOW.
4. Optional tempo:
   - TEMPO : QUANT — detected notes wait for the next grid tick and play together, so a scene falls onto a beat.
   - TEMPO : SEQ — detections are played back one note per tick. SEQ : LOOP appends them to a looping step sequence: empty steps fill first, then the oldest step is overwritten, so the phrase keeps following the scene; LEN sets the loop length. SEQ : ONCE queues them instead: each tick plays the oldest one and it is gone, so a burst becomes a run spread over time and the line goes silent when the queue empties. CLEAR empties either.
   - BPM taps through 80, 100, 120, 140, 160; drag it sideways for any value between 40 and 240. DIV is the tick: 1/4, 1/8, 1/16.
   - The square at the end of the TEMPO line is the click: it fills on every tick and grows on the downbeat (each quarter note). Notes only ever sound on those fills.
5. Open the ≡ button at the frame's top-right for scale, root, synth, sensitivity, camera and flips. The camera keeps running underneath.

Tuning tip: if nothing fires, raise SENS (drag the track right). If the wall texture itself fires, lower it. LAST shows the strength of the last trigger, so you can see how far above or below threshold a scene sits.

CONTROLS

Three caption lines under the frame are the performance controls; tap a row to change it. Everything else lives in the settings sheet behind ≡.

WHERE	ROW	VALUES
screen	MODE	dynamic / static
	PLAY	stopped / playing (static)
	SWEEP	4s / 8s / 16s per pass (static)
	LINE	vertical / horizontal
	LOW	bottom / top, or left / right: pitch direction, also sweep direction
	TEMPO	off / quant / seq
	BPM	tap 80…160, drag 40…240 (tempo on)
	DIV	1/4, 1/8, 1/16 (tempo on)
	SEQ	loop / once (seq)
	LEN	8 / 16 / 32 steps (seq loop)
	CLEAR	empty the sequence or queue (seq)
settings	SCALE	penta minor, penta major, minor, major, blues, chromatic
	ROOT	C … B (octave 3)
	SYNTH	sine, triangle, saw, square, supersaw
	SENS	0.25 LOW … 4.00 MAX; tap cycles presets, drag the track below it for any value
	CAMERA	cycle devices: front / back, USB cameras, browser inputs
	FLIP H	mirror the picture and the detection together
	FLIP V	same, vertically
	LAST	intensity of the last trigger
	FULLSCREEN	web only
	CLOSE	back to the picture

Selfie cameras (laptop, phone front) start with FLIP H on so the picture behaves like a mirror.

DESIGN

Monochrome: white ground, black ink, one typeface (IBM Plex Mono). The picture is the camera’s luminance lifted towards paper. The passepartout is whatever the image does not cover, so it adapts to every camera aspect and screen. The grid is anchored at the image centre with 12 cells across the short side on every device; the dead centre of the camera is always a grid crossing.

HOW IT WORKS

What you see is what is analysed. Each platform delivers raw frames; the app never shows a platform camera preview.

```
platform frame — orient(rotation, flipH, flipV) ─▶ Gray (screen-oriented luminance)
                                                   ├─▶ drawn as the picture
                                                   └─▶ stripOf(line, band 10 px)
                                                          └─▶ detector ─▶ bins ─▶ notes
```

- lib/feeds.dart — camera sources. Mobile uses the camera plugin’s image stream (no preview widget), macOS keeps the camera_macos view at one invisible pixel because

it only streams frames while its texture is painted, web samples a detached <video> through a canvas (lib/web_feed.dart).
- lib/camera_strip.dart — orient rotates and flips into Gray; stripOf averages a thin band along the line into a 1-D strip cropped to the whole centred cells; rgbaOf makes the display pixels.
- lib/canvas_screen.dart — geometry (Geom: cell size, bins, snapping, fit), the frame painter, caption rows, settings sheet, tempo clock.
- lib/motion_detector.dart — two detectors over the strip. MotionDetector (dynamic) diffs consecutive strips per bin with an EMA, Schmitt trigger, two-frame confirmation, refractory period and rejection of global motion (camera shake). StaticDetector (static) scores each bin by the contrast of its brightest quarter against the strip median and fires on the rising edge, with no global rejection, so a row of spots is a chord.
- lib/sequence.dart — the looping step sequence for TEMPO : SEQ.
- lib/scale_mapper.dart — scales, bin → MIDI note, tick length.
- lib/sampler.dart — SoLoud waveform voices, one source per (instrument, pitch). Pitch is set with the waveform frequency, because SoLoud's synth derives its phase step from the voice sample rate and play speed cannot retune it. Sample-based instruments can plug in behind the same cache later.

Bins along the line are the whole centred grid cells: 12 on the short axis, 16 on the long axis of a 4:3 camera. Pitch = root + scale interval of the bin, octaves stacking upward.

RUN AND BUILD

```
flutter pub get
flutter run -d macos          # or ios / android / chrome
flutter test                  # detectors, geometry, sequence, widget tests
flutter build web --release   # output in build/web
```

Platform notes:

- macOS — the first launch asks for camera permission. When started from a terminal the prompt is attributed to the terminal app; allow it once.
- Android / iOS — the camera permission is requested on first launch. Back camera is the default; switch with CAMERA.
- Web — browsers need a gesture before audio, hence the start screen. The camera permission prompt follows. FULLSCREEN is in the settings sheet. Audio runs single-threaded on plain static hosting; heavy UI work can crackle.

Deployment: every push to main builds the web app and publishes it to GitHub Pages through .github/workflows/pages.yml.

DOCUMENTATION PDF

docs/MotionDraw.pdf is generated from this README (pandoc → Typst). Enable the hook once with git config core.hooksPath tools/hooks; every commit that touches README.md then regenerates docs/readme-body.typ and the PDF and stages them. By hand: tools/readme_pdf.sh.

REPOSITORY

```
lib/           app code (see How it works)
test/          unit tests for orient/strip, detectors, scale, sequence; widget tests for
```

```
the UI
web/           Flutter web scaffold with the SoLoud loader scripts
assets/fonts/  IBM Plex Mono (OFL)
docs/          Typst template and generated PDF of this README
tools/         readme_pdf.sh and the pre-commit hook
android/ ios/ macos/   platform runners
```

LICENSE

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